

card reader. With the NOOBS now loaded into the microSD card, Raspberry Pi can now be setup. Insert the microSD card into the card slot below the Raspberry Pi. Plug the USB keyboard into one of the USB ports and plug the mouse into another USB port. You can also connect a Bluetooth adapter into one of the USB ports.

Turn on your monitor or TV set and make sure it is set to the proper input (e.g. HDMI 1 or Component). Plug in the HDMI or video hardware into the monitor and connect the receiving end of the wire into the Raspberry Pi unit. Connect an ethernet cable to the router beforehand if you plan on working with files on the internet or with any IoT devices. Connect the other ends of these devices to the Raspberry Pi. Alternately, connect the Wi-Fi adapter to the Raspberry Pi and attach any power supplies or ethernet cables to the system.

On attaching the power supply into the power outlet, Raspberry Pi will be turned on booted up to Raspberry Pi. A power light will glow indicating that there is a stable connection. Download the Raspbian operating system on the Raspberry Pi. Install Raspbian Pi by clicking on the setup file. The system will show a prompt asking for the installation to occur stating that the microSD card will be overwritten with Raspbian operating systems. Once installation ends, Raspbian will boot up. In order to configure any changes, click Menu in the upper left corner of the screen, go to Preferences in the dropdown menu. Select Raspberry Pi Configuration in the Preferences list. When the configuration window appears, click on the Localisation tab.

Click on Set Locale and choose your location. Click on Set timezone and choose your local time. Click on Set Keyboard and choose the keyboard language. Once the reconfiguration is done, the Raspberry Pi will require a reboot. The reboot window will appear automatically. Click Yes to continue. Once the reboot is complete, the Raspberry Pi unit will be up and ready to be used for new programs.

Getting Your Code Started

You can choose to run any program that you want. Let's start with a simple command-line output program where Raspberry Pi will greet the user on starting up.

Download the Espeak package to make the Raspberry Pi speak "Welcome to the world of Robots". If the package isn't installed on the system, run the following command in the terminal to install it:-

- `sudo apt-get install espeak`